

A Report on my findings on the analysis of the healthcare dataset

Analysis Questions

1. Patient Demographics & Profile

- What is the age distribution of patients in the hospital dataset?

Ans : The dataset consists of people across all ages with the elderly(66+) taking the bulk of the dataset, The elderly(66+) makes up approximately 29% of the whole dataset while the children and teenager(0-18) barely makes up 2% of the dataset.

- How does patient distribution vary by gender?

The hospitals had more male patients generally, but the difference between the male population and the female population is not so much.

- Are certain blood types more common among patients with specific conditions?

Every blood type was affected by one disease or the other but there were blood types that were more susceptible to some diseases than others. For example AB+ and A- are more susceptible to hypertension (2% chance higher) than others while B+, O+, and AB- are more susceptible to Arthritis than others.

2. Medical Conditions & Diagnosis Trends

- What are the most common medical conditions recorded?

The most common medical conditions were Diabetes, Arthritis and Cancer. They were the top 3 medical conditions respectively.

- Which medical conditions are most frequent across different age groups?

The children & Teenagers (0-18) were more prone to Obesity than others, this can be due to the fact children tend to eat a lot of junks.

The elderly (66+) and the middle ages(51-65) are the most prone to hypertension.

The older youths (36-50) and the middle ages (51-65) were seen to be more prone to diabetes.

The youths(19-35), middle ages(51-65) and elderly(66+) were seen to be more prone to cancer.

The children, the youths and the elderly were more prone to asthma.

The older youths(36-50), middle ages(52-65) and the elderlies(66+) were more prone to arthritis.

- Is there any relationship between blood type and specific medical conditions?
Yes, though not very strong, some blood types had approximately 2% chance of getting some diseases more than others.

3. Admissions & Discharge Patterns

- What is the trend of patient admissions over time (by date/month)?
By the end of 2019, number of patient admissions were around the range of 7000-7500, but the number peaked by the year 2020 to a little above 11000 and stayed in that range to 2023. This was probably to the global pandemic that struck during that time. There was a massive decline in 2024 to below the 4000 levels.
- Which admission type (emergency, elective, etc.) is most common?
The elective admission type was the most common, that means most of the diagnosed medical conditions were not life threatening yet.
- What is the average length of hospital stay per condition or admission type?
The average length of stay irrespective of admission type was approximately at 15 days

4. Financial & Insurance Analysis

- What is the total and average billing amount per condition?
Obesity, Diabetes and Asthma were the most expensive medical condition per patient

while Diabetes, Obesity and Arthritis made the most money for the hospitals overall due to the high number of patients suffering from those medical conditions.

- Which insurance providers cover the highest number of patients?
Cigna and Medicare are the two leading medical insurance providers.
- Is there a relationship between admission type and billing amount?
There was no significant relationship between admission type and billing amount.

5. Hospital & Doctor Performance

- Which hospitals handle the highest number of cases?
LLC Smith and LTD Smith are the hospitals with the most influx of patients
- Which doctors manage the most patients or highest caseloads?
Michael Smith, John Smith and Robert Smith handled the most caseloads.
- Are there differences in outcomes across hospitals or doctors?
Yes, different doctors were called to different hospitals so it was not much of a monopoly of a doctor working across different hospitals.

6. Treatment & Outcomes

- What is the distribution of test results (normal, abnormal, inconclusive)?
The bulk of the test results were abnormal which may mean that most of the conditions of the patients became worse
- Which medical conditions have the highest recovery (normal test results) rate?
Asthma and Hypertension come out top as the medical conditions with the highest recovery rate.
- What medications are most commonly prescribed for each condition?
Lipitor was the most prescribed medication for both diabetes and cancer
Ibuprofen and Lipitor were the most prescribed for hypertension.

Aspirin was the most prescribed for Arthritis while paracetamol was the most prescribed for Asthma.

7. Operational Insights

- Are there peak periods for hospital admissions?
The peak periods were around the third quarters of the years i.e the summer months(June, July, August)
- Which patient groups (age, condition, admission type) require the most care resources?
From observation, The elderlies and the youths require the most care resources, They constitute the most of the population of patients admission and they are also the age groups that are more prone to diseases like cancer.
- Are there patterns between room allocation and patient outcomes?
Unfortunately, the relationship was not very significant(A correlation of 0.0002).

8. Future Trends & Predictive Insights

- Based on historical data, what is the projected trend of patient admissions in the coming months?
Based on historical data, we would see patient admissions peaking at the summer months, the end of the years and the beginning of the years.
- Which medical conditions are likely to increase in frequency over time?
I see Diabetes, Arthritis and Cancer increasing over time based on historical data.
- What patient demographics are most at risk of frequent hospital admission in the future?
The elderlies and the youths.
- Which conditions are likely to result in longer hospital stays based on past patterns?
Asthma, Arthritis and cancer.

- What early indicators (age, condition, admission type) can help predict high billing cases?

Medical Condition appeared to be the strongest indicators that would help predict high billing cases.

- Which hospitals or departments may face higher patient loads in the future?
LLC Smith is expected to experience higher influx of patients and most of them would be diabetic patients and patients suffering from Arthritis and Cancer.

- What insights can be used to improve resource planning (staff, beds, medication stock)?

From my observation, the hospital should focus getting stronger medications for the patients that will help cope with their medical conditions.

The hospitals should also employ more doctors and provide more beds in their facilities to reduce the overload of patients in one or two hospitals.